Module 9 · Lecture 2

Bayesian Optimization on a Bosch Etch Twin

Eight knobs, one a whole number; five specs; BO, random search and an LLM agent

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What you'll take away

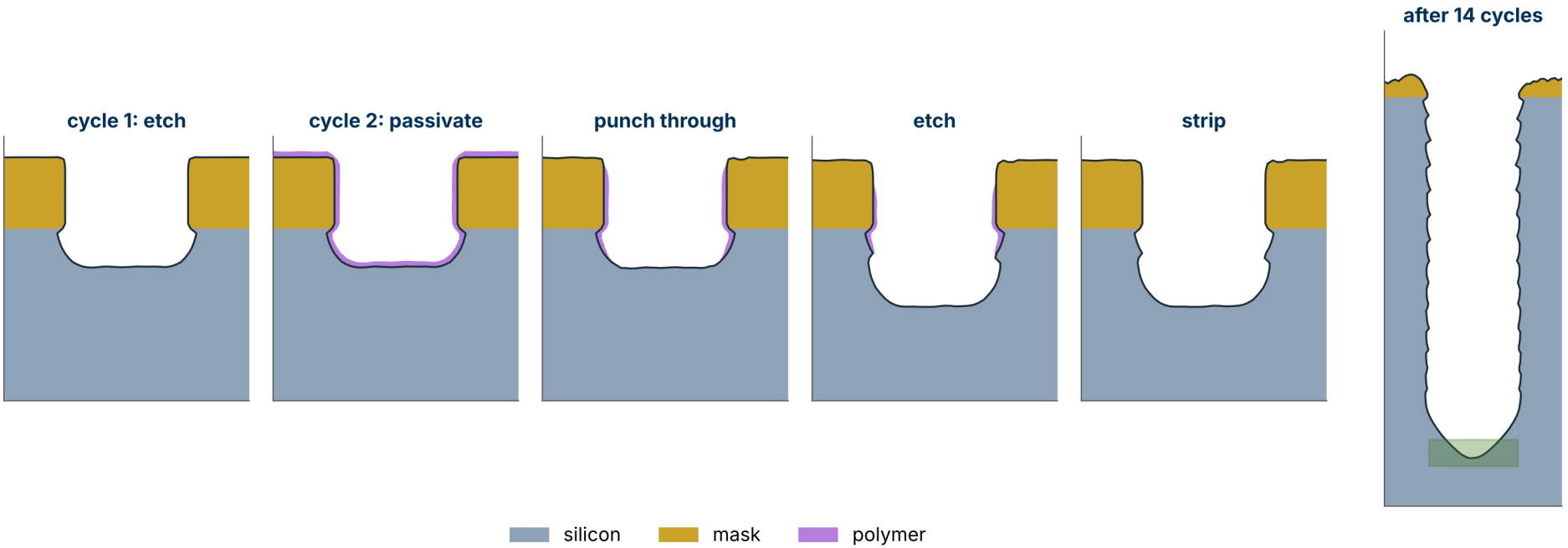
1. **The Bosch process as a twin:** deposit, punch through, etch, strip — repeated — and the five numbers a cross-section is judged by.
2. **A whole-number knob:** two ways to give a GP the cycle count, and what the difference cost here.
3. **The race, measured:** BO, structured BO, random search and five runs of an LLM agent, every reported recipe re-run.
4. **Profiles throughout:** every search shown as the trenches it produced.
5. **Throughput against the mask:** what a fast recipe spends.

A

THE BOSCH TWIN

Deep, straight trenches one cycle at a time

The Bosch process, one cycle at a time



Lärmer & Schilp, Robert Bosch GmbH: US patent 5,501,893 (priority 1992, granted 1996). Alternating etch and polymer steps make the deep silicon trenches behind MEMS and through-silicon vias.

Takeaway Each etch step scoops the sidewall; the polymer protects what is already etched. This recipe: 14 cycles to 4.06 µm.

Eight knobs and five specs

Knobs

cycles: 4–20, whole number

etch time: 0.5–3 s per cycle

passivation time: 0.3–2 s (0.05 µm of polymer per s)

polymer sticking: 0.002–0.2: low coats deep, high coats near the top

SF_6 rate: 0.05–0.5 µm/s, isotropic

ion rate: 0.02–0.3 µm/s, directional; erodes the mask

ion exponent: 50–1000: higher is more directional

breakthrough: 0.5–2× the polymer thickness

Spec

depth: 4.00 ± 0.15 µm

scallop: ≤ 50 nm

sidewall angle: within ±1°

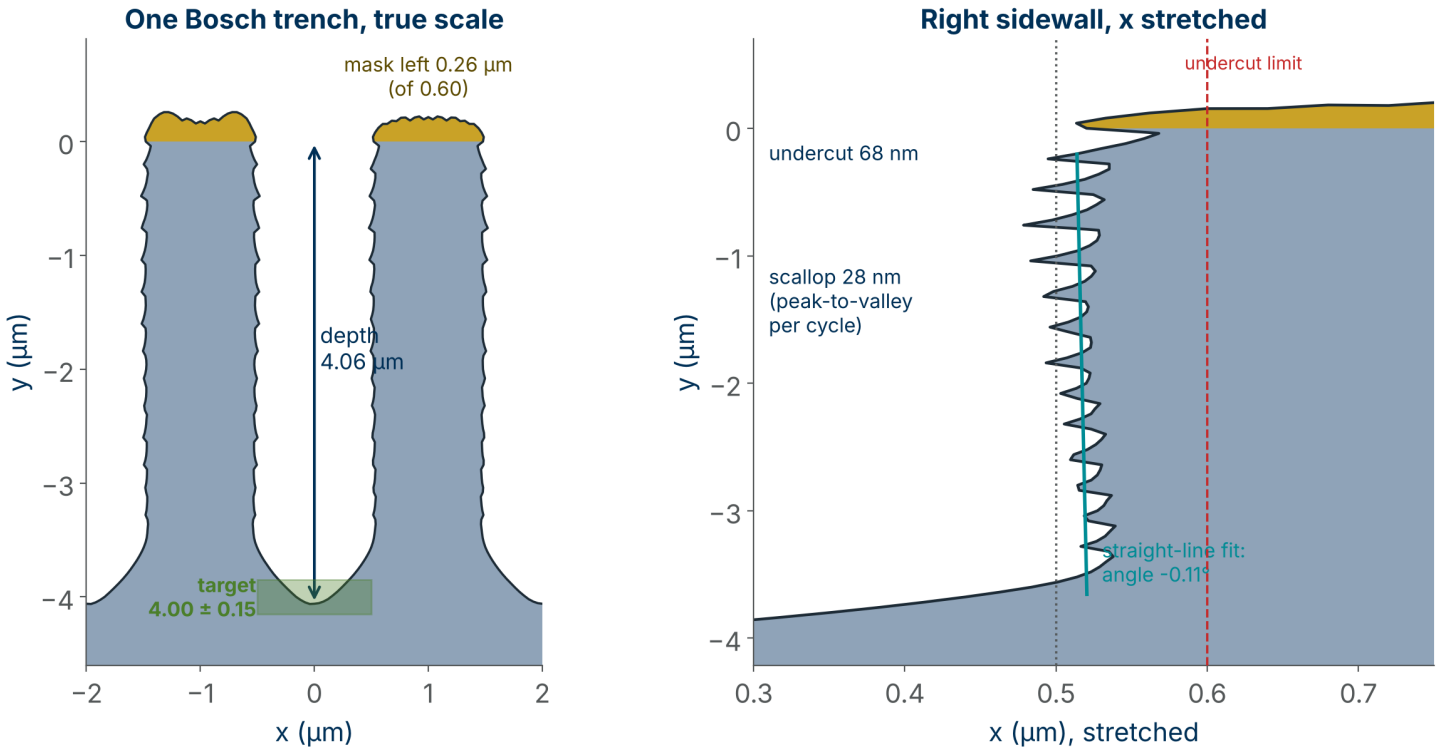
undercut: ≤ 100 nm

mask left: ≥ 0.25 of 0.60 µm

Loss: five penalties added; a pass needs all five in spec. Built on ViennaPS 4.6.2's own Bosch example; about 5–15 s per run.

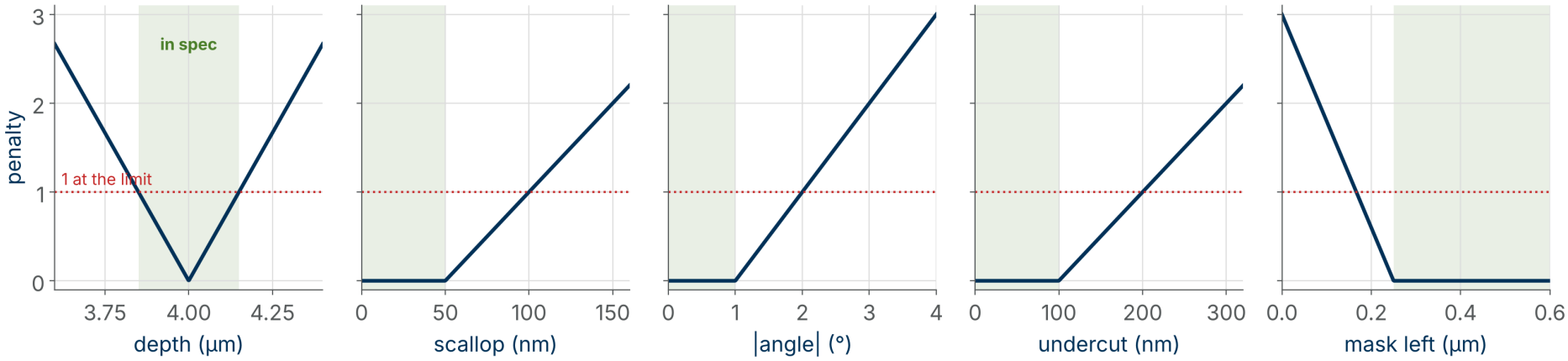
Takeaway Twice the knobs of the SF_6/O_2 twin, and the cycle count only takes whole values.

What the metrology reads off a Bosch profile



Takeaway This run: depth 4.06 µm, scallop 28 nm, angle -0.1°, undercut 68 nm, mask left 0.26 µm.

The loss: five specs, one number



loss = depth + scallop + angle + undercut + mask penalties

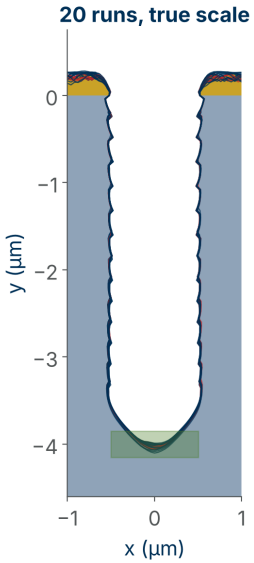
Each penalty is 1 at its limit (mask: 1 at 0.17 µm left). **Pass: depth penalty ≤ 1 and the other four = 0.**

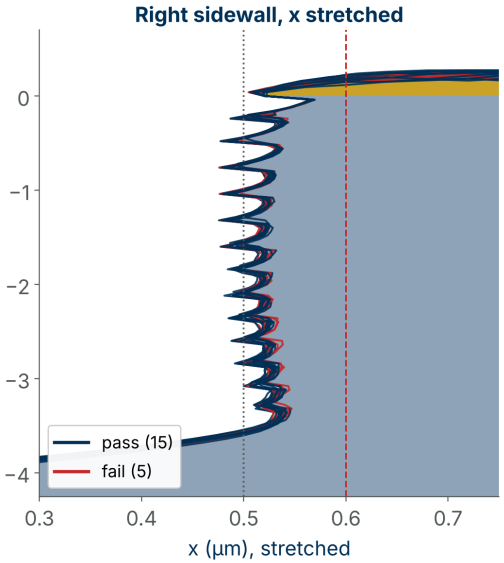
Pass with loss 0.41: depth 4.062 µm; the whole loss is the depth term.

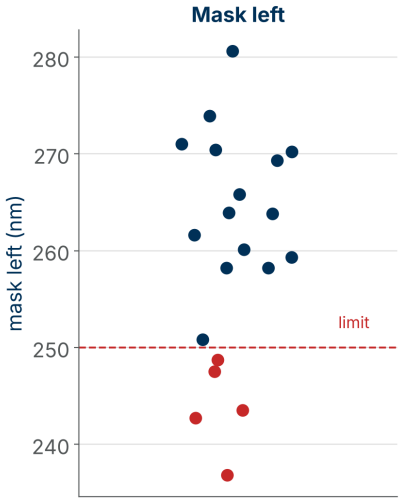
Fail with loss 0.098: angle 1.09°, just past 1°.

Takeaway A loss below 1 is not a pass: only depth may be off target inside spec.

The same recipe, twenty times







Run-to-run spread

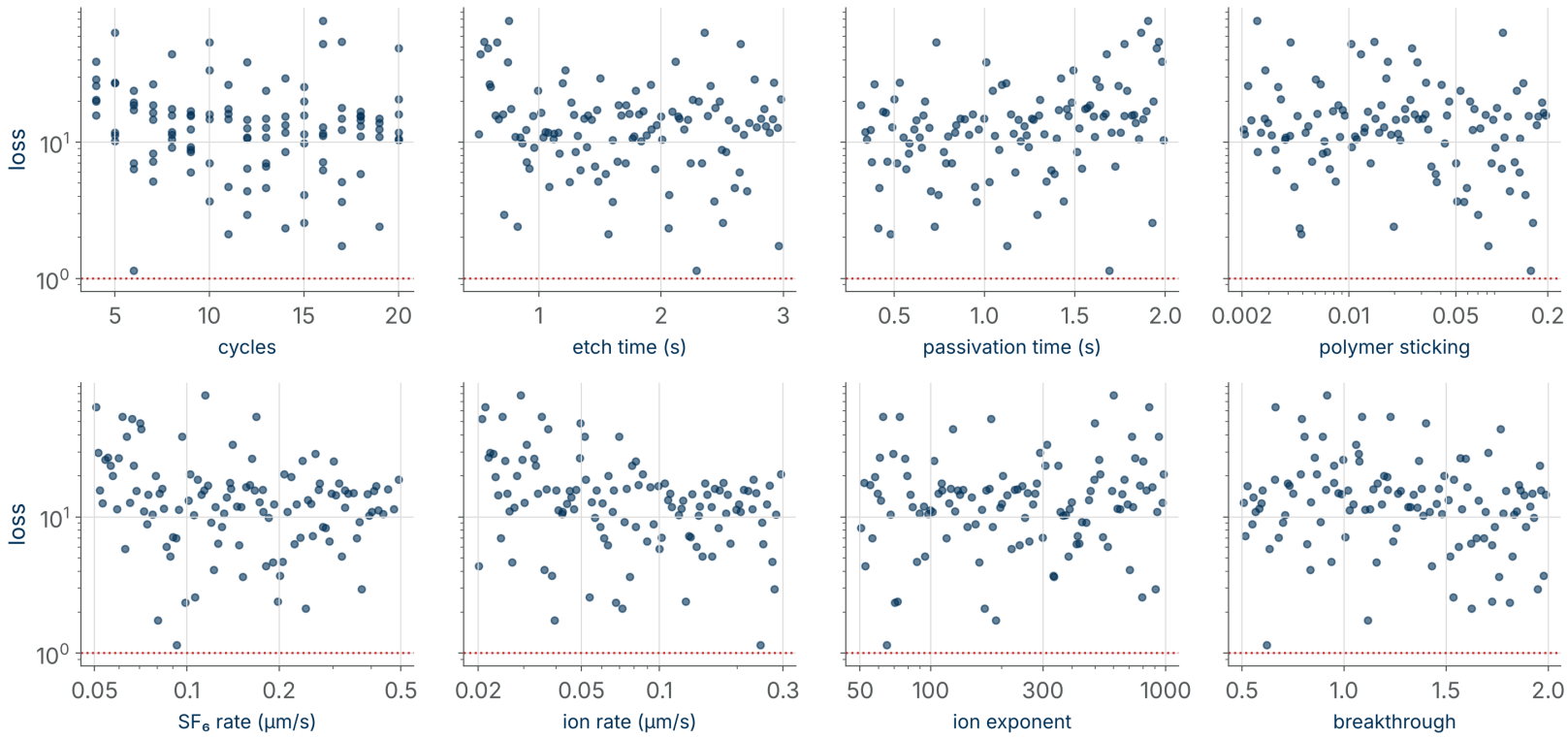
15 of 20 runs pass.

SD: depth 28.9 nm, scallop 2 nm, angle 0.08°, undercut 1.1 nm, mask 11.7 nm.

Failures by term: mask 5

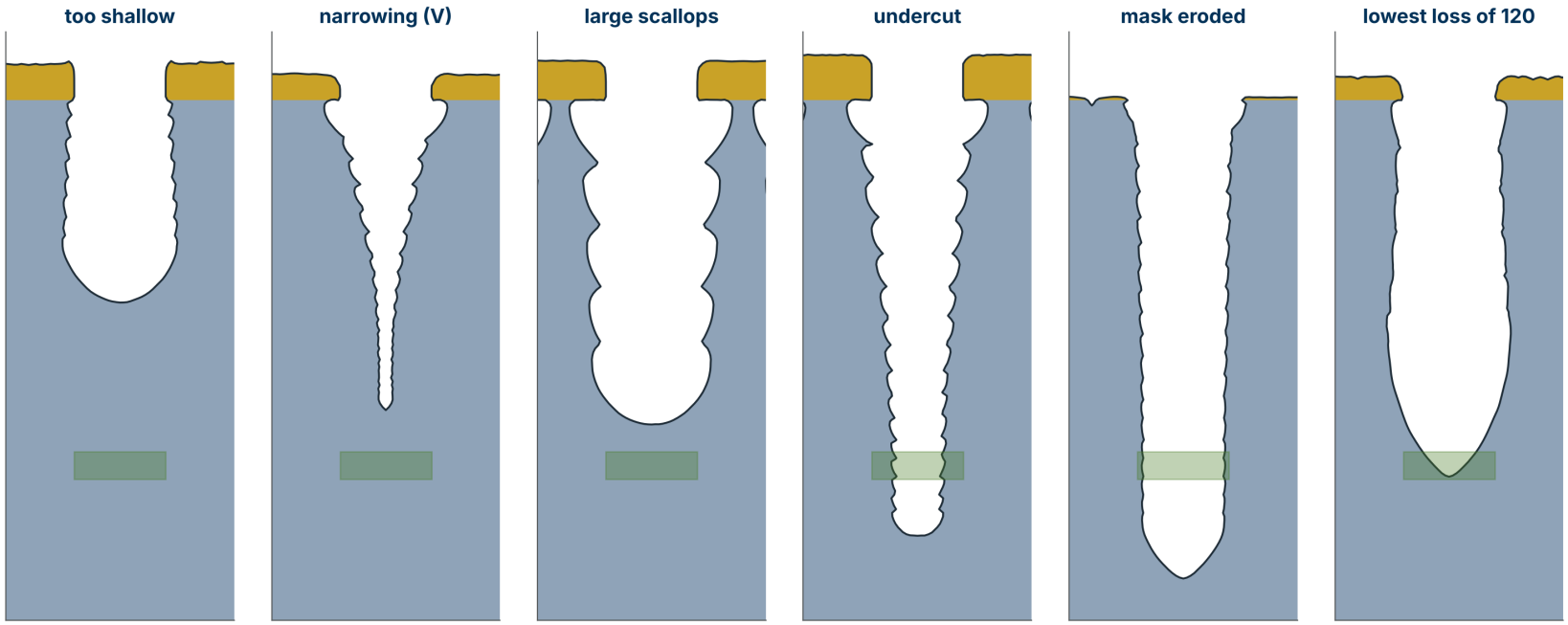
Takeaway A single run is one draw. Every reported recipe in this lecture is re-run before it counts.

The landscape: 120 recipes, none in spec



Takeaway Best loss 1.15. Share in spec, term by term: depth 3%, scallop 58%, angle 32%, undercut 45%, mask 81%.

The landscape as cross-sections



Takeaway Too shallow, narrowing, rough, undercut, mask gone: the same eight knobs make all of them.

B

THE RACE

BO, structured BO, random search and five agent runs

A GP with a whole-number knob

1

Relax, then round

Treat cycles as continuous; round the proposal before running it. The model believes 9.3 cycles exists, and the acquisition can rank 9.4 above 9.0 although both run as 9.

2

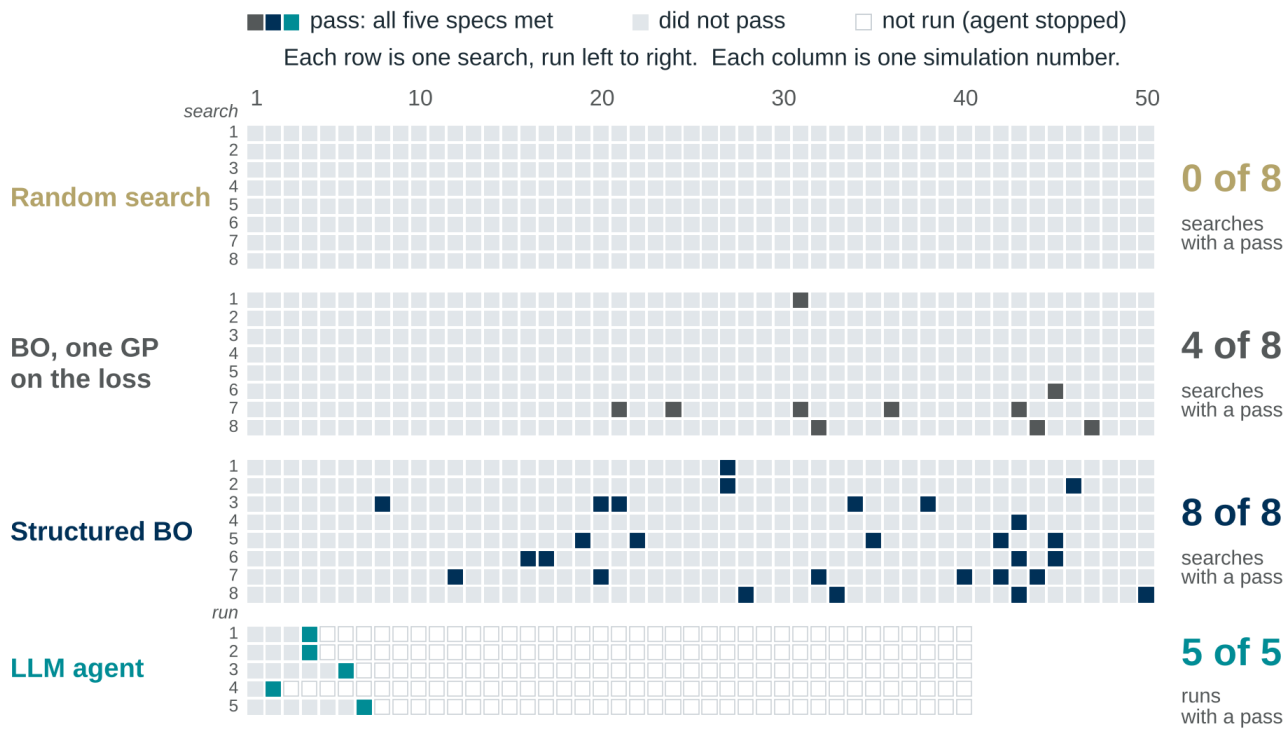
Round inside the GP

BoTorch's Round input transform rounds cycles before the kernel sees them (Garrido-Merchán & Hernández-Lobato, 2020). The model is flat between whole numbers.

Common to every BO search: unit-cube inputs, log axes for the four rate, sticking and exponent knobs; noisy expected improvement in log form; 5 Sobol starts; 50 simulations.
Structured BO: one GP per metric — depth, scallop, angle, undercut, mask left — with the loss computed from their samples inside the acquisition, as in the SF_6/O_2 lecture.

Takeaway BO, structured BO and random search: 8 searches each. Relax-and-round: 5.

Reading the results: “4 of 8” and “5 of 5”



How to read “4 of 8”

Pass: one simulation that meets all five specs at once.

Search: one complete campaign from a fresh start, 50 simulations long: one engineer, 50 wafers. Each row is one search; random search and both BO versions ran 8 each.

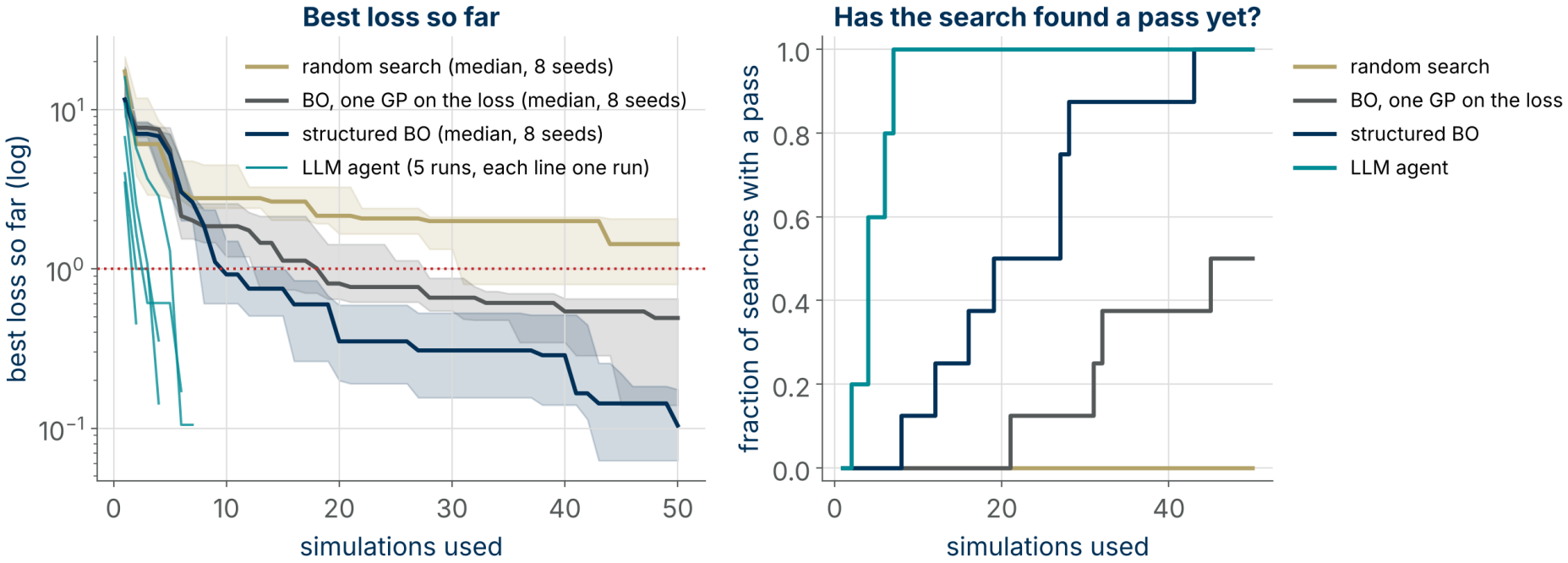
4 of 8: 4 of the 8 BO searches had at least one pass. The other 4 never passed.

5 of 5: all 5 agent runs passed. This five counts runs. Each run stops at its first pass.

First pass: the simulation number of a search's first filled square.

Takeaway “4 of 8” counts searches: 4 of 8 BO searches found at least one recipe meeting all five specs.

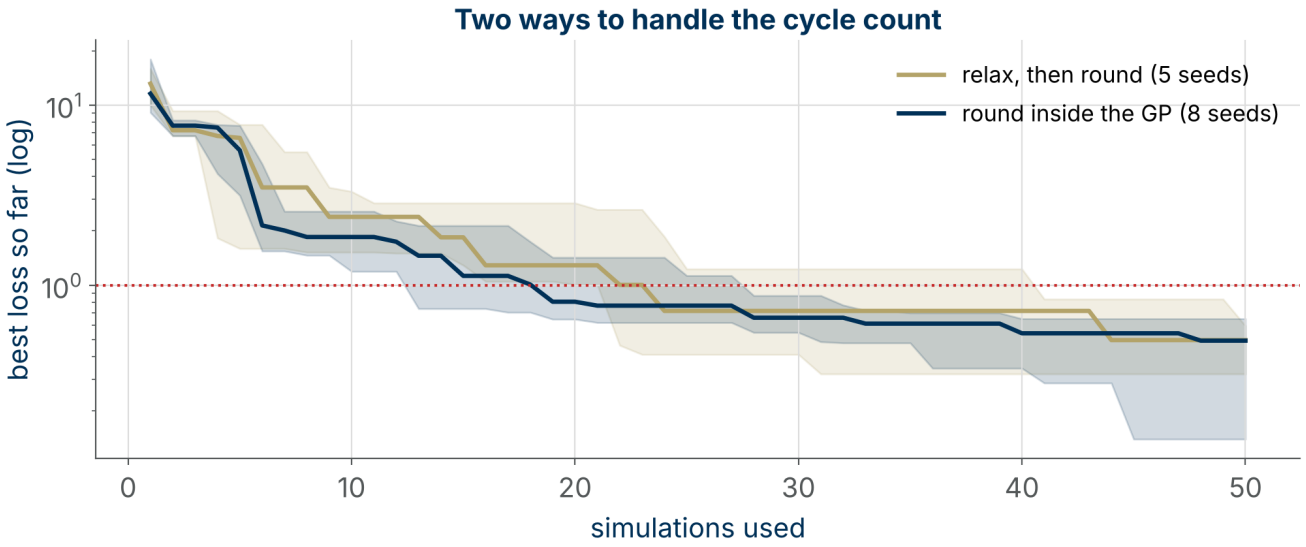
The race



Median best loss at 50: structured BO 0.10, BO 0.49, random 1.4. Median first pass: agent 4, structured BO 23.

Takeaway Searches that observed a pass: structured BO 8 of 8, BO 4 of 8, agent 5 of 5, random 0 of 8.

The cycle count: round inside the GP, or round afterwards?



What it changed here

Searches with a pass: round inside 4 of 8, relax 2 of 5.

Median best loss at 50: 0.49 for both.

Confirmed recipes: 2 of 8 and 1 of 5.

With 8 and 5 searches, the two are indistinguishable.

Takeaway Here the integer handling mattered less than the model of the loss: structured BO passed in all eight searches.

The agent: the agent-lab loop on the Bosch twin

1

Same loop

claude-sonnet-5-5; one tool that runs a recipe and returns metrology. The tool text states each knob's physical effect; BO gets only the ranges.

2

Bigger budget

Up to 40 simulations per run, against 12 in the agent lab. Every run stopped at its first pass, after 4, 4, 6, 2, 7.

3

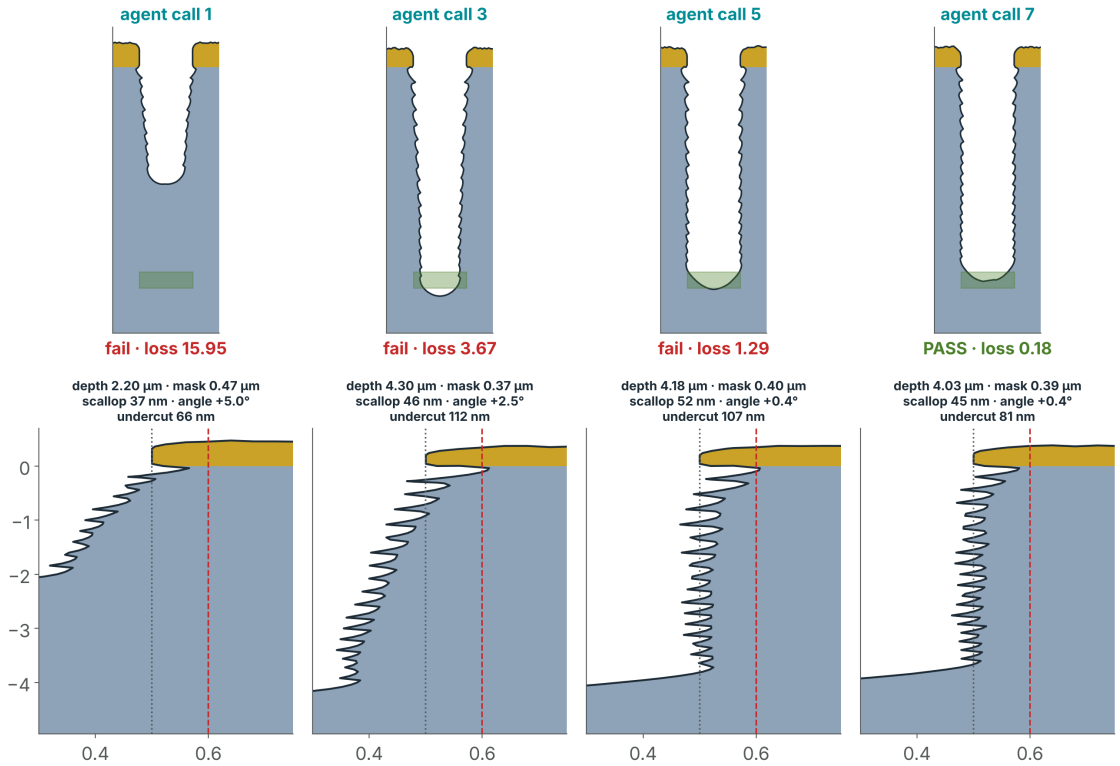
Five runs

Five independent runs instead of one; 23 simulations in total.

Agent run 5, before call 7: *"Everything passes except the angle, which is 1.02° against a limit of 1.0°. Raising sticking has reduced the taper so far, so I'll nudge it from 0.1 to 0.14."*

Takeaway Agent: a pass in 5 of 5 runs, median first pass 4. Structured BO: median 23.

One agent run, seen as profiles



Agent run 5: what changed

Call 1: baseline, 10 cycles

loss 15.95

Call 3: cycles 10→18; SF_6 rate 0.15→0.25

loss 3.67

Call 5: cycles 18→17; sticking 0.02→0.1

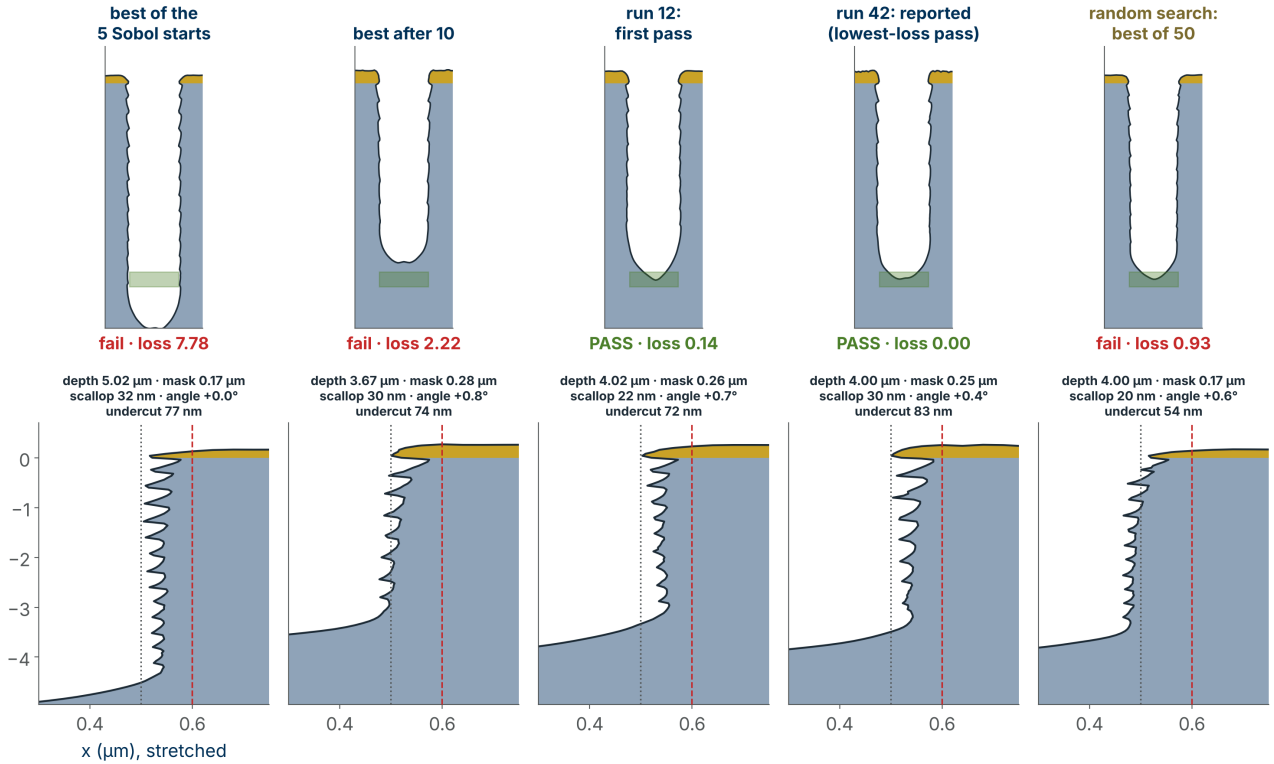
loss 1.29

Call 7: cycles 17→19; etch time 1→0.85; sticking 0.1→0.14

loss 0.18, pass

Takeaway Each profile is the recipe the agent chose at that call, from its own log.

One structured-BO search, seen as profiles



Structured BO, search 7 of 8

Run 4: loss 7.78.

Run 8: loss 2.22.

Run 12: loss 0.14, pass.

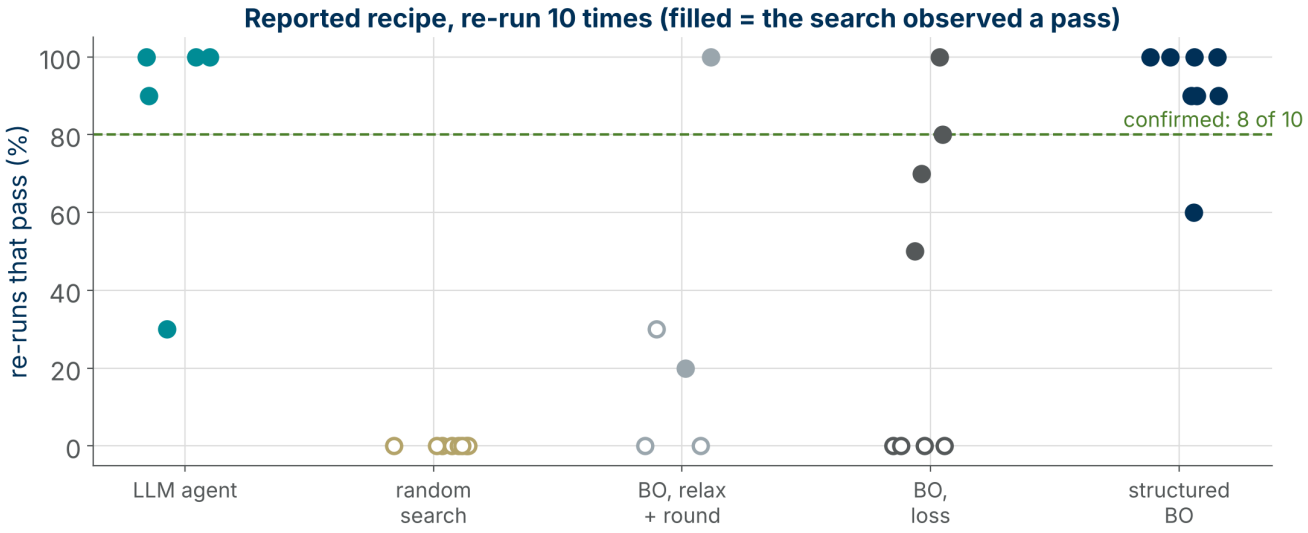
Run 42: loss 0.00, pass.

Reported recipe: 10 of 10 re-runs pass.

Random search, same seed: best of 50 is on depth and walls but leaves 0.17 µm of mask.

Takeaway Run 42: depth 4.00 µm, scallop 30 nm, angle 0.4°, mask 0.25 µm, in 22 s.

Does the reported recipe pass again?



Confirmed: 8 of 10

Each search reports its best passing recipe, or its lowest loss if it never passed. Re-run 10 times.

Agent: 4 of 5 confirmed

Random: 0 of 8 confirmed

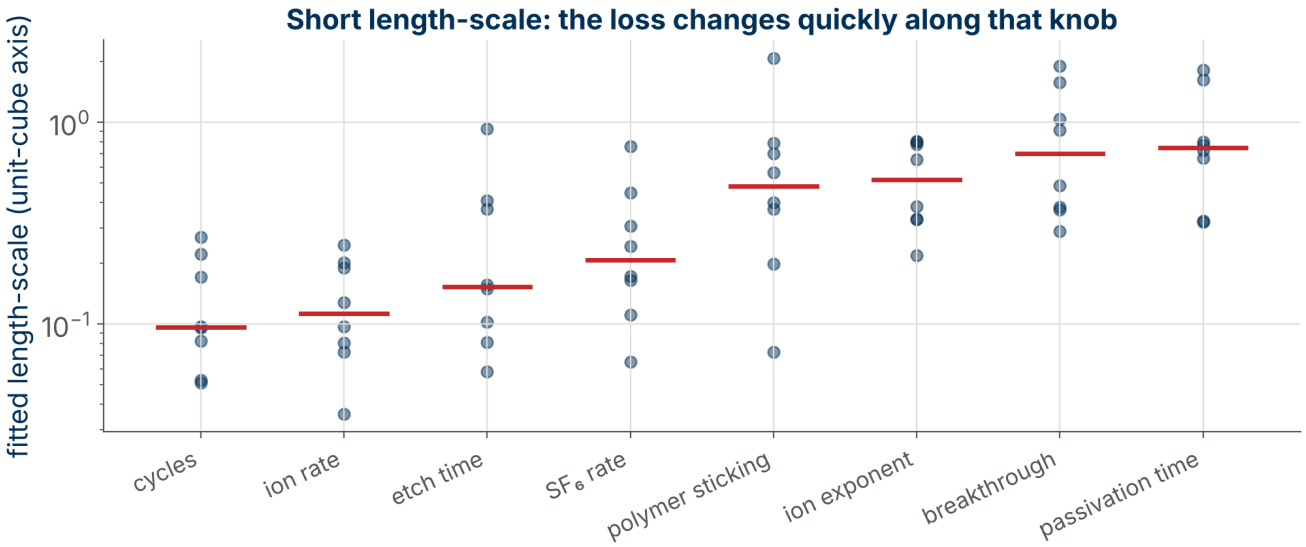
BO, relax: 1 of 5 confirmed

BO: 2 of 8 confirmed

Structured BO: 7 of 8 confirmed

Takeaway Confirmed recipes: structured BO 7 of 8, agent 4 of 5, plain BO 2 of 8.

What the model learned about the knobs



Read it as sensitivity

One fitted length-scale per knob, from the plain-BO searches. Short means the loss changes quickly along that knob.

Most sensitive: cycles, ion rate, etch time.

Least: passivation time.

Cycles, etch time and ion rate set how deep each cycle goes and how much mask it costs.

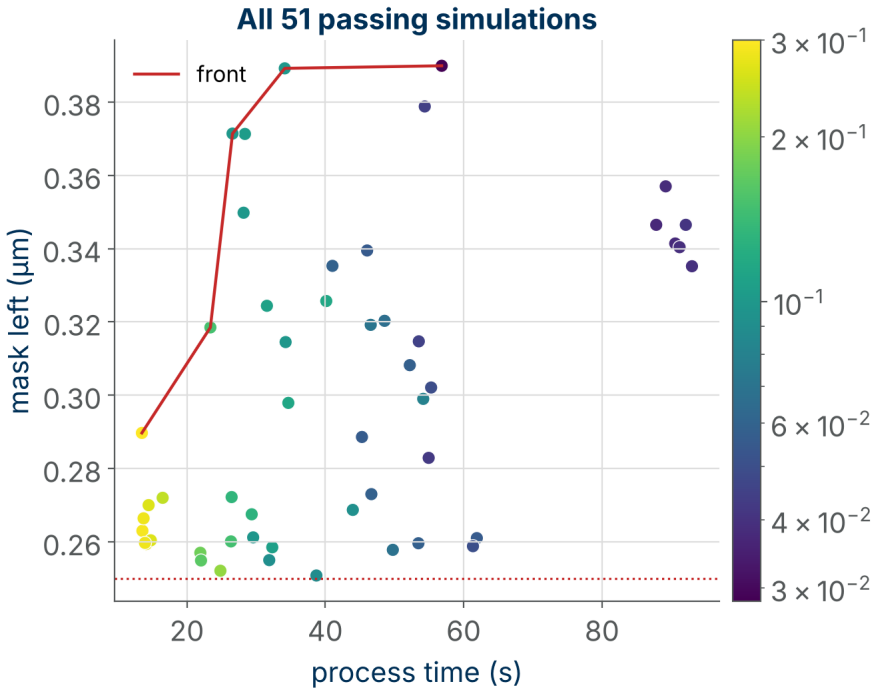
Takeaway The fitted kernel is a rough sensitivity ranking. Check it against the physics of each cycle.

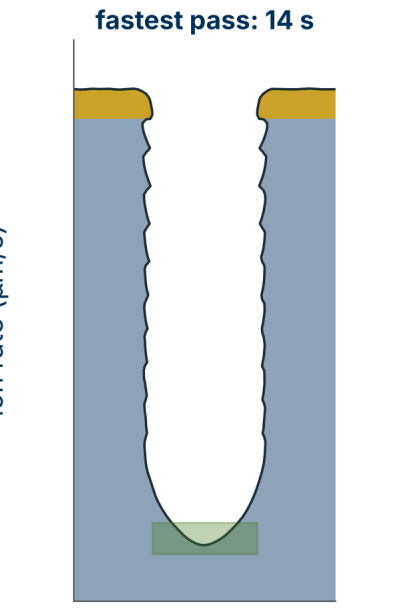
C

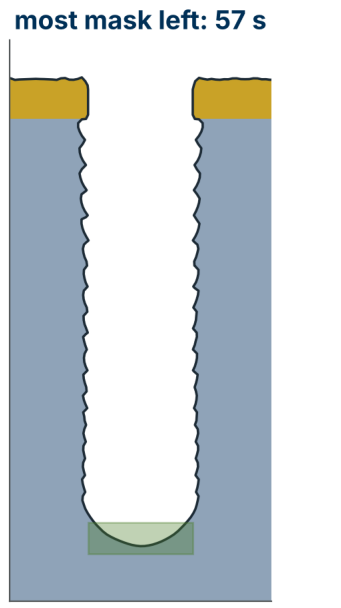
THROUGHPUT AGAINST THE MASK

What a fast recipe spends

Faster etch, thinner mask







Takeaway 51 passing simulations span 14–93 s. The fastest leans on ions and keeps 0.29 µm of mask; the one with the most mask takes 57 s.

D

WHAT IT MEANS

The scoreboard, and where each method fits

The scoreboard

Method	Searches	Observed a pass	Median first pass	Confirmed (8 of 10)
Random search	8	0 of 8	—	0 of 8
BO, relax and round	5	2 of 5	23.5	1 of 5
BO, one GP on the loss	8	4 of 8	31.5	2 of 8
Structured BO	8	8 of 8	23	7 of 8
LLM agent	5	5 of 5	4	4 of 5

Median first pass is taken over the searches that observed a pass.

Takeaway Budget 50 per search; agent up to 40, stopping at its first pass.

Where BO and the agent fit

1 Reading the metrology

Where it wrote reasoning (12 of 18 changes), it named a cause, such as more sticking for taper. Median first pass 4; structured BO 23.

2 What each method was told

The agent's tool description explains what each knob does physically. BO gets only the knob ranges and learns every effect from data.

3 How the loss is modelled

One GP per metric: passed 8 of 8, confirmed 7 of 8. One GP on the loss: 4 of 8 and 2 of 8.

4 Re-running the answer

Confirmed: agent 4 of 5, structured BO 7 of 8, indistinguishable at these counts. One agent recipe passed 3 of 10; every failure was on the angle.

Takeaway With physics in its tool and in its training, the agent passed sooner; both needed re-runs.

Takeaways

1. **Model what you measure.** One GP per metric: 8 of 8 passed; one GP on the loss: 4 of 8.

2. **Re-run before you report.** A passing recipe passed 15 of 20 re-runs; one agent recipe, 3 of 10.

3. **Physics in the tool helps.** With each knob's effect written down, the agent passed in a median of 4 simulations; structured BO took 23.

4. **The loss only asks for what it contains.** Passing recipes differed 7× in process time; the fast ones spend the mask.

Discussion: *which Bosch knobs would you hand to BO, which to an agent with the physics written down, and which would you fix from experience?*

Exercises (in the notebook)

1 Integer handling

Fix cycles at 10 and re-run relax and round-inside. How much of the gap is the integer knob?

2 Throughput as an objective

Constrained multi-objective BO (qLogNEHVI) on process time and mask left. Compare its front with the scatter.

3 Agent interface

Strip the causal hints from the agent's tool description. How many more calls, over five runs?

4 Warm start

Seed structured BO with each agent run's first two calls. How much sooner does it pass?

5 Robust recipe

Optimize the chance of passing 8 of 10 re-runs instead of the loss.

6 Mask selectivity

Set MASK_ION_FRAC to 0.05 and 0.2. How does the best recipe change, and why?

Every figure rebuilds from data/ in about a minute; the job scripts in the README re-run the simulations. The agent runs need an Anthropic API key.

Georgia Tech

Related work & where to read next

The Bosch process	Lärmer & Schilp, "Method of anisotropically etching silicon," US 5,501,893, Robert Bosch GmbH (priority 1992, granted 1996).
Integer inputs in BO	Garrido-Merchán & Hernández-Lobato, "Dealing with categorical and integer-valued variables in Bayesian optimization with Gaussian processes," Neurocomputing (2020).
BoTorch	Balandat et al., NeurIPS (2020); Ament et al., "Unexpected Improvements to Expected Improvement for Bayesian Optimization," NeurIPS (2023).
The twin	Reiter & Filipovic, "ViennaPS: A flexible framework for semiconductor process simulation," SoftwareX (2025).
Human + machine	Kanarik et al., "Human–machine collaboration for improving semiconductor process development," Nature 616, 707–711 (2023).